

Karthik Reddy Yalala

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Education

Bachelor of Science in Computer Science

Arizona State University, Tempe, Arizona

May 2027

(GPA 4.00)

Relevant Coursework: Data Structures, Algorithms, OOP, Operating Systems, Software Engineering, Artificial Intelligence, Machine Learning, Cybersecurity

Honors and Certifications: Dean's List: Fall 2023–Spring 2026; **AWS Certified Cloud Practitioner**

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C, C++, SQL, HTML, CSS

AI Engineering: Amazon Bedrock, Bedrock AgentCore, LangGraph, LangChain, CrewAI, MCP, RAG, Agentic AI, Tool Use, Claude, OpenAI APIs, Hugging Face

LLMOps and Evaluation: Embeddings, Hybrid Search, Vector Search, Reranking, Structured Outputs, Guardrails, Prompt Safety, Adversarial Testing, RAG Evaluation, Regression Testing, CloudWatch, Cost and Latency Monitoring, Feedback Analytics

Backend, Cloud and DevOps: FastAPI, Node.js, Express, REST APIs, WebSockets, SSE, AWS Lambda, API Gateway, DynamoDB, S3, CloudFront, Cognito, Textract, Docker, Git, Continuous Integration and Continuous Delivery, AWS CDK

Databases: PostgreSQL, MongoDB, MySQL, Redis, Pinecone, Firebase

Experience

ProfessorDM Technologies Inc.

Remote

LLMOps and AI Infrastructure Engineer

Aug 2026 – Present

- Lead technical architecture and service boundaries for a **7-person engineering team**, coordinating cross-service integration for a production AI MVP serving a client with **165K+ Instagram followers**.
- Design **LLMOps evaluation gates and AI guardrails** with the AI Lead, establishing prompt and model change controls, automated validation, and production-safe failure handling.
- Own **AI infrastructure, cloud deployment, and security architecture**, including CI and CD pipelines, role-based access control, secrets management, and integrated development and staging environments.
- Establish production **observability and release readiness** through application logging, error tracking, telemetry metrics, and an automated regression testing strategy for upcoming beta trials.

Avis Budget Group (Fortune 500)

Parsippany, NJ

AI Enablement Intern

Jun 2026 – Aug 2026

- **Architected and engineered 4 production GenAI applications** across API design, full-stack implementation, evaluation, and production handoff; saved **170+ engineering and product hours**, contributed to solutions projected to deliver **\$1M+ in annual savings** at customer scale, and presented results directly to the Director, VP of Customer Experience, and CEO.
- Shipped a production **RAG search service** on AWS using Lambda, API Gateway, DynamoDB, and CloudFront; integrated **Amazon Titan V2 embeddings** for multilingual retrieval across **100+ languages**, alongside metadata-filtered retrieval, exact and semantic caching, Bedrock Guardrails, and regression evaluation.
- Replaced a third-party **LLM SaaS** with an in-house **AI assistant on Bedrock AgentCore** to keep compliance and employee data internal; engineered **SSE streaming, Azure AD SSO and MFA**, and prompt safety scanning, with a custom adversarial test suite covering hallucination and safety failure cases.
- Engineered a **self-correcting NL-to-SQL agent loop** using Vercel AI SDK over Bedrock Claude to inspect scoped schemas, generate and repair SQL, and return tables, summaries, and charts; excluded PII at the schema layer and saved **70 hours**.

Projects

Crucible AI - AI Interview Platform – LangGraph, Amazon Bedrock, FastAPI, React, DynamoDB, AWS Lambda 2026

- Built and deployed a production **AI interview platform** with a **6-agent LangGraph pipeline** for intake, planning, live interview execution, evaluation, cross-session memory, and coaching; rewrote each session's question plan around persisted candidate weak spots and verified adaptation through end-to-end tests on generated prompt text.
- Designed a **golden-dataset evaluation harness** with **35 regression and adversarial cases** including buzzword-padded inputs and subtle logic traps; achieved **95%+ routing and action accuracy** across live Bedrock test runs by tiering Claude Haiku for extraction and Claude Sonnet for multi-step reasoning.
- Shipped to AWS Lambda, API Gateway, DynamoDB, Cognito, and CloudFront with **100+ backend tests** and CI; eliminated concurrent-write memory loss using optimistic locking and enforced **schema-validated structured outputs** so no raw model text crosses agent boundaries.

FinRAG MCP - Financial Intelligence RAG Server – Python, AWS Bedrock, MCP SDK, FastAPI, Pinecone, Ragas 2026

- Built a production **MCP server** for cited financial intelligence over **500+ SEC filings** across **50 companies** with EDGAR ingestion, table-aware extraction, hierarchical chunking, and weekly refresh.
- Designed a **four-stage retrieval pipeline** with query rewriting, hybrid BM25 and dense search, reranking, and numerical grounding verification; benchmarked against a **300-question FinanceBench and custom evaluation dataset** with Ragas regression gates.
- Deployed AWS serverless infrastructure with Cognito OAuth 2.1 and PKCE, model-tier routing, CloudWatch observability, and a dashboard tracking **faithfulness, numerical accuracy, and cost per query**.